Mesoscopic Community Based Portfolio Optimization: A comparison test

Billy Jun Ming Lim
billy.ljm@gmail.com

Manuel Luci
manueluci.wqu@gmail.com

Source Code available on Github:
https://github.com/BillyLjm/Mesoscopic-Community-Portfolio-Optimisation

ABSTRACT

Accurate estimation of the stocks correlations is key for portfolio optimisation, yet empirical correlation matrices are often noisy especially for larger number of assets. We investigate the use of network analysis and clustering techniques – including Louvain, Label Propagation, DBSCAN, and Agglomerative Clustering – to identify stable mesoscopic correlation structures in the S&P 500. Portfolios created based on these techniques are then compared with traditional portfolios, demonstrating that our methods more reliably predicted future volatilities across long ten-year time frames, although the realised volatilities were comparable to traditional techniques. Thus, our techniques might be suitable for investors who want to have a clearer idea of their long term portfolio performance.

1 Introduction

One of the key tenets of portfolio optimization is the benefit of diversification: combining assets that are imperfectly correlated with each other can allow investors to achieve higher returns with lower risk since the assets' individual returns sum but the risks partially offset each other. However, this requires the accurate estimation of the correlation matrix between the assets, which becomes increasingly difficult with a large number of assets. For example, the Marcenko–Pastur distribution shows that random noise in the individual asset returns will still contribute some structure to the observed correlation matrix; and the correlation matrix can also change over time such as most assets becoming correlated during periods of financial crisis. Thus, there has been a growing interest in the last few decades to study the stability of the market's correlation structure, so that models can be developed to account for those effects. In this work, we aim to similarly refine the estimation of the covariance matrix by studying a newly proposed method of using network analysis techniques to elucidate the mesoscopic covariance structure between assets. This mesoscopic covariance should be more stable over time, allowing us to construct optimised portfolios which perform better out-of-sample and over longer time horizons.

2 Theoretical Framework

Markowitz's Modern Portfolio Theory (1952) is the most widely-used portfolio building frameworks and has served as the benchmark for more advanced algorithms [1]. At its core, it seeks to balance the trade-off between the expected return and the volatility of the portfolio; usually leveraging on the anti-correlation between the constituent assets to minimise the overall volatility of the portfolio. However, it assumes that the correlation matrix between the assets can be accurately measured via historical price movements, and this simplistic method has since repeatedly been shown to be inaccurate.

Perhaps most influentially, Laloux, Cizeau, Bouchaud, and Potters (1999) showed that empirical correlation matrices contain significant amounts of random noise, much of which would still be present even if the matrix was completely random [2]. Specifically, they showed that the eigenvalue spectrum of large empirical correlation matrices exhibit a hump at small eigenvalues, which was consistent with that of a random matrix given by the Marcenko–Pastur distribution. Thus, they proposed that we should not blindly use the empirical correlation matrices for risk management, and instead filter the random noise along with any other corrections to determine the true correlation between financial assets.

Building on Laloux et al.'s seminal work, Tola, Lillo, Gallegati, and Mantegna (2008) proposed the use of agglomerative clustering to elucidate the true correlation matrix instead [3]. In their work, they used single linkage clustering with an ultra-metric distance derived from the empirical correlations, and they showed that the resulting correlation matrices were more reliable than Laloux et al.'s method, with the predicted correlation matrices being more consistent with the realized one.

And since then, there has been numerous similar research which applied various clustering and network analysis techniques in portfolio construction, such as DBSCAN, label propagation algorithm, and K-means clustering [4, 5, 6]. In 2015, Yang et al. analysed the diffusion of information across the financial community on Twitter to underline the influence of these communities on the volatility and short-term returns of assets [7]. In the same year, Almog et al. showed how the binary information given by the sign of network returns enough information to describe the mesoscopic structure of the network [8]. In 2021, Anagnostous et al. applied community detection algorithm to analyse the credit default swap market, and showed that the communities are more heterogeneous across sectors, but more geographically coherent [9]. In 2022, Wu et al. proposed a K-means cluster portfolio based on pattern and market trend to identify possible regime changes and build an automatic trading strategy [5]. In 2023, Kamuhanda et al. deployed community detection to identify possibly-illegal Bitcoin transaction [10].

And most notably for our work, Zema, Fagiolo, Squartini, and Garlaschelli (2024) proposed first filtering the random noise as proposed by Laloux et al., and then clustering the filtered correlation matrix via the Louvain method [11]. They argued that the correlation between clusters represents a broader, mesoscopic structure of the markets which should be more stable and persist for longer periods. Thus, they proposed that we should allocate equally within the clusters and only optimize our allocations between these mesoscopic clusters, showing that this method was more reliable than the empirical correlation matrix and an equal weight portfolio.

Our work builds upon these studies by investigating whether modern techniques can extract the true correlation matrix of the market more reliably. In particular, we follow the approach of Zema et al., applying it across a range of clustering methods, including agglomerative clustering, DBSCAN, Louvain method, and label propagation algorithm, to benchmark their performance against one another as well as against Markowitz's minimum-variance portfolio. This allows for a more comprehensive comparison of how different clustering techniques uncover the underlying structure of financial markets.

3 Methodology

For this paper, we consider the subset of S&P 500 stocks that are in the index today, and that have complete price information within our time horizon of 2000 to 2024. This selection yielded an investment universe of 349 stocks, whose price data is readily accessible and which should still be broad enough for mesoscopic structures required by our algorithm to emerge. For the analysis, we adopt a methodology similar to to Zema et al. [11]. The main steps are:

1. Remove the market component and random noise from the covariance matrix
2. Identify the remaining mesoscopic communities
3. Optimise the weights between clusters to form the portfolio
4. Benchmark the out-of-sample performance of the portfolio

3.1 Market and Random Noise

Following Laloux et al., we first remove the market and random noise and from our correlation matrix [2]. The market component is the largest eigenvalue $\lambda_{\max}$ of the correlation matrix, and it is attributed to the common dependence of all assets on global market events. Since it is unavoidable, Laloux recommended the removal of this market component. As for the random noise, it is predicted to have an eigenspectrum given by the Marcenko-Pastur distribution $f(\lambda)$ of

$$f(\lambda) = \frac{\kappa}{2\pi\sigma^2}\frac{\sqrt{(\lambda-\lambda_+)(\lambda-\lambda_-)}}{\lambda}$$

where the λ_- and λ_+ are the minimum and maximum eigenvalues of the distribution respectively, given by

$$\lambda_{\pm} = \sigma^2\left(1 \pm \sqrt{1/\kappa}\right)^2$$

And since we are working on correlation matrices, we have $\sigma^2 = 1 - \lambda_{\max}$ and $\kappa = T/N$, where T the number of timestamps of the price information and N is the number of assets. For simplicity, we will ignore the exact shape of the Marcenko-Pastur distribution, and instead filter out eigenvalues of the correlation matrix which are within the maximum λ_+ and minimum λ_-. We can then rebuild the correlation and covariance matrices based on the remaining unfiltered eigenvalues and eigenvectors, which should reflect the mesoscopic structure of our investment universe more accurately.

3.2 Mesoscopic Communities

Next, we group the stocks into communities based on the remaining mesoscopic correlation matrix, following the approach proposed by Tola et al. who employed agglomerative clustering, and Zema et al. who used the Louvain method [3, 11]. However, our work extends on these studies by directly comparing the performance of these algorithms, and also including more modern algorithms such as DBSCAN from clustering analysis, and Label propagation from network analysis.

3.2.1 Agglomerative Clustering

In agglomerative clustering, the assets are recursively grouped into clusters based on a distance metric [12]. The algorithm works in a bottom-up fashion, where each asset starts out in their own individual cluster, and at each step the closest two clusters are merged until a stopping criterion is met. This is a standard technique, and in our implementation we used the version provided in the scikit-learn package [13].

For our use case, the chosen distance metric is one minus the correlation between the assets; so that perfectly correlated assets will have a distance metric of zero, and anti-correlated assets will have a distance metric of 2. Additionally, following the findings of Tola et al., we used single linkage where the distance between clusters is set to be the minimum distance between the constituent assets [3]. Finally, we stopped the agglomerative clustering when 5 clusters have been reached, since that is the typical number of communities identified by the non-parametric network analysis methods as we will show later.

3.2.2 DBSCAN

DBSCAN (Density-based spatial clustering of applications with noise) clusters assets based on the density of their neighbourhoods, as measured via a distance metric [14]. Assets are separated into core points which have a high density of other assets in their neighbourhood, border points which are in the neighbourhood of core points, and noise which are not in the neighbourhood of any core points. Core and border points in the neighbourhoods of each other are then clustered into the same community, while noise are left unclustered.

We used the implementation provided by scikit-learn with a distance metric of one minus the correlation between the assets, for reasons identical to agglomerative clustering described before [13]. As for defining the neighbourhood, we set a threshold distance of 0.86, since it yielded 4 clusters which again is similar to the non-parametric network analysis methods shown later.

3.2.3 Louvain Method

The Louvain method detects communities in a network by maximising the modularity $Q(c)$ of some partition c of the assets into communities [15, 16]. This modularity essentially quantifies the weight of edges within the same cluster, and is mathematically given by

$$Q(c) = \frac{1}{2m} \sum_{i,j} \left[w_{ij} - \frac{(\sum_k w_{ik})(\sum_l w_{lj})}{2m} \right] \delta(c_i, c_j)$$

where w_{ij} is the weight of the edge between asset i and j, $m = \sum_{x,y} w_{xy}$ is a normalisation factor of the total weight of all edges, c_i denotes the community asset i is in, and $\delta(c_i, c_j)$ is the Kronecker-delta function which yields 1 if assets i and j and in the same community, or 0 otherwise.

The resulting optimised partition c identifies the mesoscopic communities of the stocks, such that edges with higher weights tend to be contained within the same community. Zema et al. used these communities to optimize their portfolios and benchmark portfolio performance, and in our work we replicated their work with the implementation provided by NetworkX [11, 17].

3.2.4 Label Propagation Algorithm

Lastly, the label propagation algorithm initialises each asset as a unique community label, and then repeatedly updates this label stochastically based on that of the neighbouring nodes in the network [18,19]. This propagates the labels throughout the network, and within each densely connected subnetwork, one label is likely to become dominant, thereby identifying the communities of interest. For this paper, we opted to use the version of the algorithm provided in NetworkX [17]

3.3 Portfolio Optimisation

To form the optimised portfolios, we followed Markowitz's modern portfolio theory to obtain the portfolios with minimum volatility. However, we only optimised the weights between clusters, since only they exhibit the mesoscopic covariance structure of the market which is expected to be more stable over time. The structure within each cluster is expected to be more volatile and thus we chose to allocate equal weight to every asset within the same cluster to minimise the effect of the within-cluster-volatility. These can be summed up by the following optimisation equation:

$$\begin{aligned}
&\min_{\omega} \omega^T \Sigma \omega \\
&\sum_i \omega_i = 1 \\
&\omega_i = \omega_j \ \forall \ i, j \in c
\end{aligned}$$

where ω_i is the weights of asset i, Σ is the covariance matrix, and c the community identified in the previous step.

3.4 Performance Benchmarking

To benchmark the performance of the resulting portfolios, we used a anchored walk-forward method across our time horizon of 2020-2024 [20]. We chose a 3-year burn-in period, and tried both a 1-year and 10-year testing period, since the former would yield more iterations for more robust aggregate performance metrics while the latter would test the stability of the predictions over longer periods. The main metrics we measure are the Sharpe ratio and the reliability metric.

The Sharpe ratio $\mathcal{S}$ measures the excess returns of the portfolio normalised to its volatility, with a higher ratio being more desirable since it indicates higher returns with lower volatility.

$$\mathcal{S} = \frac{R_p - R_f}{\sigma_p}$$

where R_p is the portfolio return, R_f is the risk-free return, and σ_p is the portfolio volatility.

The reliability metric $\mathcal{R}$ measures the difference between the predicted in-sample and actual out-of-sample volatility, indicating how reliable our filtered correlation matrix was over time.

$$\mathcal{R} = \frac{|\sigma_p^r - \sigma_p|}{\sigma_p}$$

where σ_p is the predicted/in-sample volatility, and σ_p^r is the realised out-of-sample volatility.

Additionally, we will also benchmark our portfolios against the global minimum variance portfolio which Markowitz's vanilla modern portfolio theory recommends, and the equal-weight portfolio which is also popular for it simplicity and remarkable historical performance [1,21,22,23].

4 Results

4.1 Market and Random Noise

Figure 1 illustrates the decomposition of the empirical correlation matrix into the market component, random noise, and mesoscopic structure as explained in section 3.1. It confirms that our empirical correlation matrix is consistent with the findings by Laloux et al., and supports the application of their filtering [2]. Specifically, our correlation matrix exhibits a peak at small eigenvalues (shown in green) which is consistent with random noise present in any matrix, and it exhibits a prominently large eigenvalue (shown in blue) similar to what Laloux et al. observed and attributed to market components.

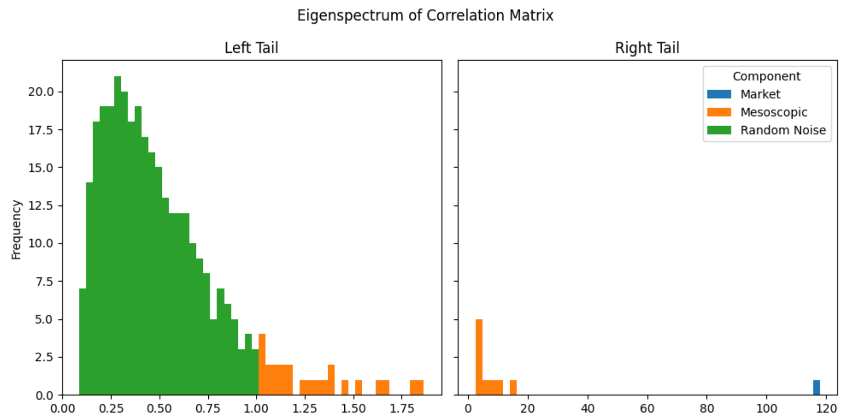


Fig. 1: Eigenspectrum of the correlation matrix for the left (L) and right (R) tail, with the attributed component

Additionally, figure 2 shows the estimated time evolution of these components, depicting the fraction of total risk/volatility attributable to each component. These were computed from daily prices using a rolling one-year window (which empirically produced stable volatility estimates) and plotted for our entire time horizon from 2000 to 2024 to depict their long-term evolution. It shows that the mesoscopic component is relatively stable over time, consistent with it capturing the stable underlying mesoscopic structure of the market. In contrast, the market and random noise frequently experience large jumps, especially during crisis such as the 2008 financial crisis and 2019 COVID-19 pandemic.

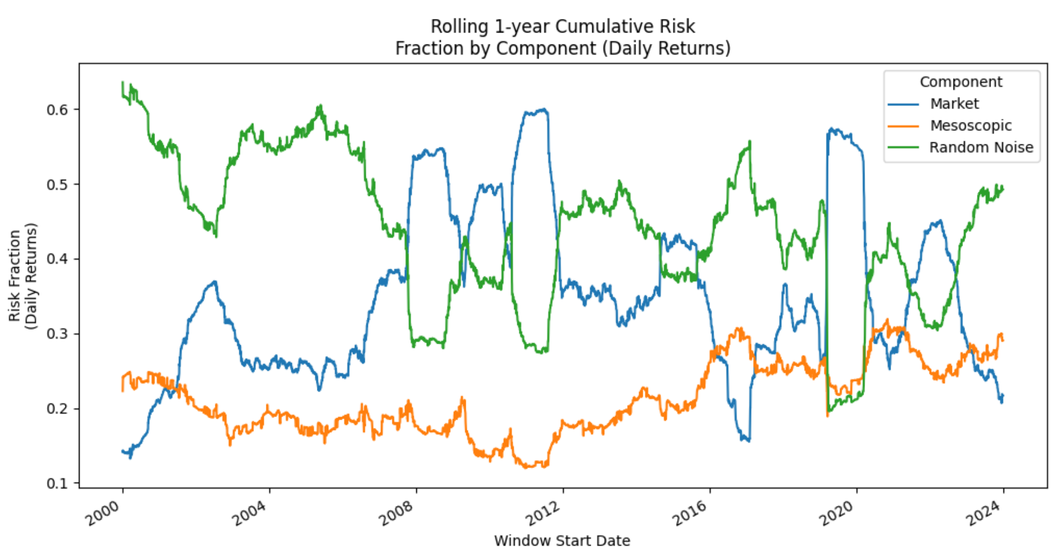


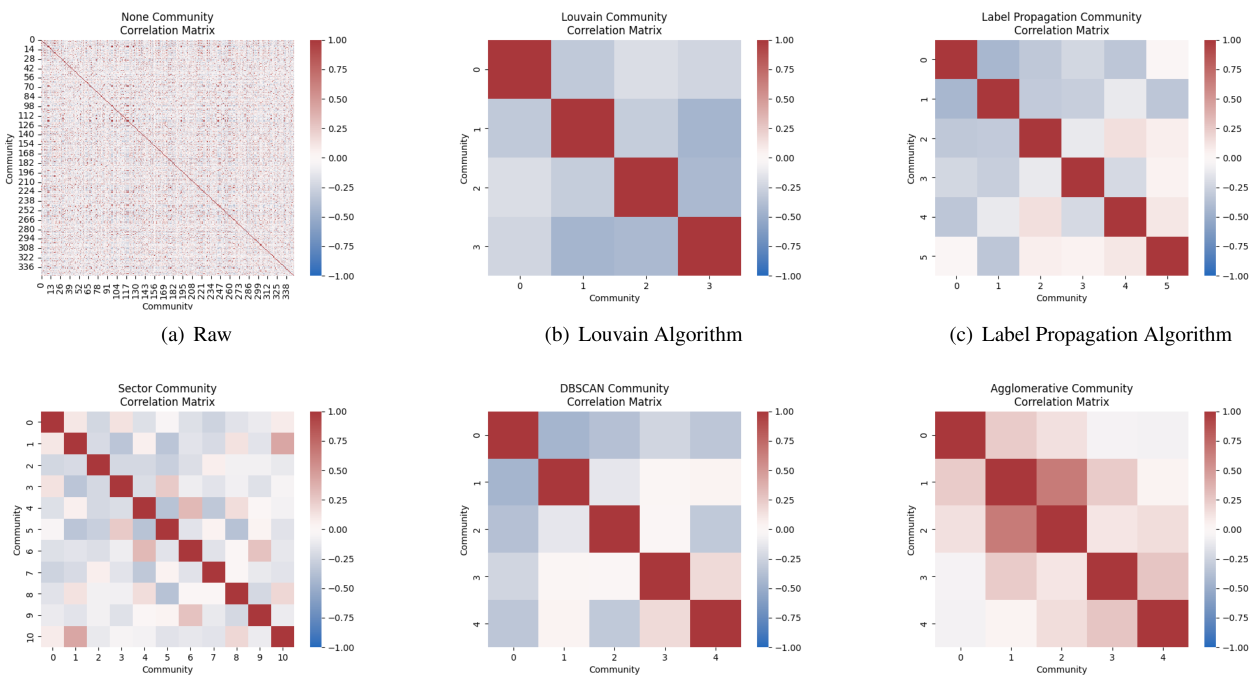
Fig. 2: Fraction of risk attributed to the market component (blue), mesoscopic structure (orange), and random noise (green).
(note: the risk/volatility was computed from daily closing prices using a rolling one-year window.)

These suggests that the mesoscopic component reflects more fundamental and longer-term structure of the market, since it stays relatively constant over longer periods of time. Thus, our subsequent analysis will focus exclusively on the mesoscopic component, since we are aiming for strategies focused on this stable structure and which ignore short-term market shocks and market-wide correlations which are unavoidable.

4.2 Community Correlation

After deriving the mesoscopic correlation matrix (as discussed in the previous section), we applied several community detection and clustering algorithms to group positively-correlated stocks into distinct communities. This is analogous to sectors in traditional finance and the motivation is that investing in these broad thematic sectors or communities should mitigate the idiosyncratic risks of investing in individual equities. We did this but simply using the negative of the correlation between equities as the distance metric which the algorithms optimises on, such that stocks with high positive correlation and likely to be in the same community.

The correlation matrix between the identified communities are plotted in figure 3. It illustrates that the community detection greatly reduces the dimensionality of the correlation matrix, from the raw matrix in figure 3(a), and simplifies the structure of the market and subsequent portfolio optimisation step.



(a) Raw (b) Louvain Algorithm (c) Label Propagation Algorithm

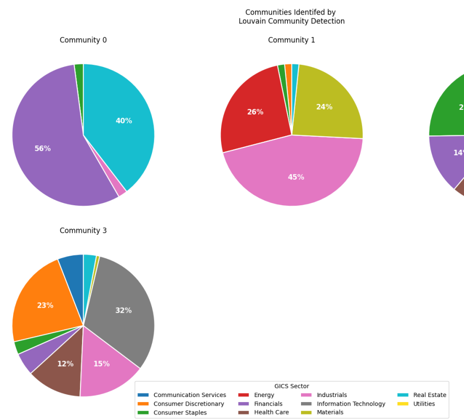
(d) GICS Sector (e) DBSCAN (f) Agglomerative Clustering

Fig. 3: Mesoscopic correlation matrix between (a) individual stocks, and communities identified by the (b) Louvain Algorithm, (c) Label Propagation Algorithm, (d) GICS Sectors, (e) DBSCAN, and (f) Agglomerative Clustering

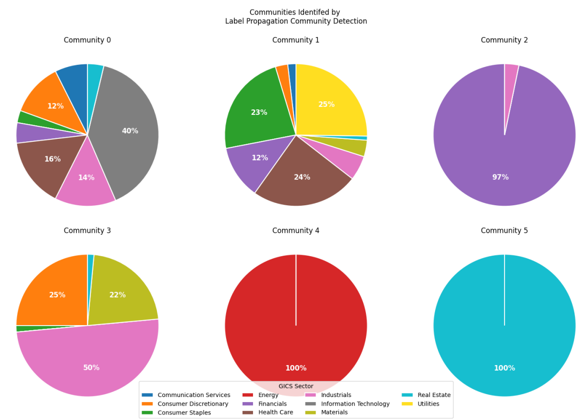
Additionally, figure 3 also gives us some indirect hints of how well each community detection algorithm performed, since the algorithms are supposed to group positively (red) correlated stocks together, leaving negative (blue) correlation between the resulting sectors. By this imprecise metric, the Louvain algorithm in figure 3(b) seemed to perform the best, identifying communities with only negative (blue) mutual correlations. Meanwhile, Label Propagation, the conventional GICS sectors, and DBSCAN (in figure 3(c), 3(d), 3(e) respectively) yielded communities with some positive (red) mutual correlations, albeit across a larger number of communities which is harder to achieve. Lastly, agglomerative clustering in figure 3(f) seemed to have failed to group positively-correlated stocks together, with all of its communities still positively-correlated with each other; this failure is also consistent with the results in the following sections.

4.3 Community Composition

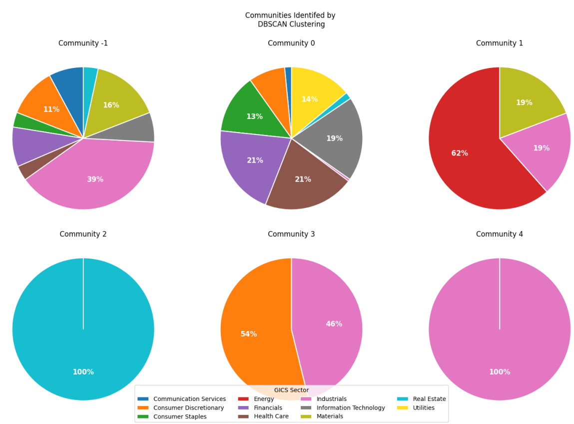
Besides comparing the correlation matrix between communities, we can also investigate the composition within each community (relative to the conventional GICS sectors) as plotted in figure 4. Again, agglomerative clustering in figure 4(d) seems to have failed, identifying trivial communities with a small number of stocks within each sector and lumping the rest in a catch-all community 1. However, broadly speaking, the results demonstrate that most of identified communities (except by agglomerative clustering) contains stocks from multiple GICS sectors, suggesting that the empirical community structure of the market (as identified via the algorithms) might extend beyond the classical GICS sector classification.



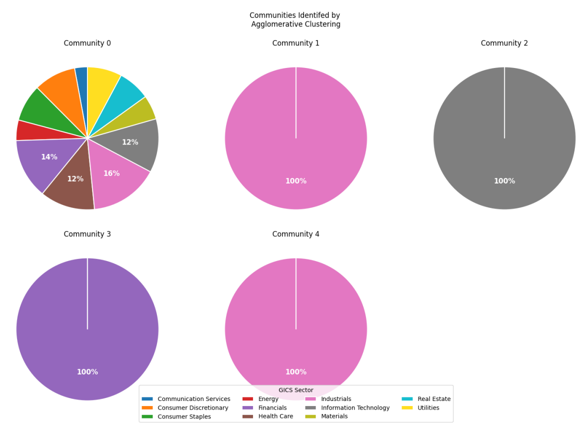
(a) Louvain Algorithm



(b) Label Propagation Algorithm



(c) DBSCAN



(d) Agglomerative Clustering

Fig. 4: GICS sector composition of the communities identified by the (a) Louvain Algorithm, (b) Label Propagation Algorithm, (c) DBSCAN, and (d) Agglomerative Clustering

We can also examine these compositions more closely to gain insight into how the GICS sectors are empirically correlated. For example, industries (pink), materials (gold), consumer discretionary (orange), and energy (red) are often grouped together in community 1 of figure 4(a), community 3 of figure 4(b), and community 1 and 3 of figure 4(c). This is consistent with established market behaviour where industrial output and consumer discretionary are positively correlated, suggesting that the community identification algorithms did find meaningful communities of stocks. Similarly, consumer staples (green), utilities (yellow), part of health care (brown), and part of financials (purple) are also often grouped together, such as in community 2 of figure 4(a), community 1 of figure 4(b), and community 0 of figure 4(c). This also aligns with market understanding where spending in these categories are stable across market cycles and thus expected to be correlated with each other. We also note that real estate (light blue) is typically grouped by itself such as in community 0 of figure 4(a), community 5 of figure 4(b), and community 2 of figure 4(c).

Additionally, the community identification algorithms identified market structure beyond the typical GICS sectors. For example, healthcare (brown) could be further separated into specific stocks that were correlated to consumer staples (green) and utilities (yellow), such as in community 2 of figure 4(a) and community 1 of figure 4(b); versus those that are more correlated to consumer discretionary (orange) and information technology (grey) such as in community 3 of figure 4(a) and community 0 of figure 4(b). Thus, these algorithms might allow community-based portfolio optimisation that are distinct and ideally better than traditional sector-based portfolio optimisation. However, these relations are more dependent on the algorithm used than the more established relations described in the previous paragraph, and thus closer study is still warranted.

4.4 Portfolio Performance

We can also back-test the portfolios constructed based on the communities identified by our algorithms versus those based on GICS sectors, individual stocks, or just blanket equal weight. Specifically, we used the past 1-year performance metrics of the communities/sectors/stocks to create global minimum variance (GMV) portfolios, which we then tested on the subsequent 10 year time period and the resulting performance metrics are listed in table 1.

Table 1: 10-year forward performance metrics of global minimum variance portfolios constructed based the communities identified by each algorithm, or GICS sectors, or individual stocks, or equal-weight stocks

Algorithm	Median Reliability	Volatility	Sharpe Ratio
Louvain	0.52	0.23	0.76
Label Propagation	0.33	0.22	0.74
DBSCAN	0.26	0.78	0.73
Agglomerative	0.08	0.25	0.71
Sector	0.17	0.20	0.82
Stocks	0.17	0.20	0.85
Equal-Weight	0.12	0.20	0.85

Based on the results listed in table 1, it is clear that the community identification algorithms used had higher median reliability of 0.30 as compared to the traditional sector, stock, and equal weight portfolios which only had 0.15 median reliability. This proves that we could more reliably predict the volatility of the portfolios created from our communities, achieving our original goal of finding a more stable mesoscopic correlation matrix from which portfolio volatility can be estimated from. Also notably, the Louvain algorithm had a significantly higher median reliability than the other algorithms, suggesting that it might be a more suitable algorithm than the others which could benefit from further investigation.

However, when comparing the more indirect but practically-important portfolio metrics such as volatility and Sharpe ratio, the benefits of the community identification disappears with the volatility being comparable (0.2) and the Sharpe ratio slightly lower (0.7 vs 0.8) as compared to traditional portfolios. Thus it would seem that the community identification methods can yield portfolios with more predictable and reliable performance which is beneficial to some investors, but this performance is not the highest among other portfolios construction techniques.

We also measured the performance of the portfolios across different time frames. In table 1 above, we focused on a more passive strategy with the portfolios being constructed and rebalanced once every 10 years. And in table 2, we measured the performance with a slightly more active strategy where the portfolios are rebalanced every year. Comparing the two tables, it is clear that our algorithms was able to reliably predict the volatility across 10years but not across 1 year, roughly bounding the timescale where our mesoscopic correlation becomes more important. However, this is exactly what we desired from our portfolio which is supposed to be focused on longer-term performance and reliability.

Table 2: 1-year forward performance metrics of global minimum variance portfolios constructed by the communities/sectors identified by each algorithm

Algorithm	Median Reliability	Volatility	Sharpe Ratio
Louvain	0.32	0.21	0.82
Label Propagation	0.31	0.21	0.78
DBSCAN	0.29	0.40	1.00
Agglomerative	0.26	0.23	0.73
Sector	0.34	0.19	0.83
Stocks	0.35	0.20	0.81
Equal-Weight	0.36	0.20	0.85

5 Conclusion

This study demonstrates how to elucidate a more stable mesoscopic correlation structure of the market, by filtering out random noise and grouping the stocks into communities. We have shown that these communities identified extend beyond traditional sectors, yielding communities that align with our traditional understanding of market structure, by also hinting at some finer structures not fully captured by traditional sectors. Additionally, we also tested the performance of portfolios constructed based on this mesoscopic structure, showing that it allowed us to more reliably predict the portfolio volatility, although the realised portfolio volatility is not lower than traditional methods. Thus, our methods are more suitable for investors that want to exactly predict the portfolio performance.

The study also demonstrates that the choice of community identification algorithm can have significant effect on the accuracy of the communities identified, with the Louvain algorithm being more suitable and agglomerative clustering failing, Thus, further research could focus on the differences between the algorithms and the resulting outcomes. Additionally, we have also highlighted early hints of more subtle correlation structures in the market, which would be interesting to study with greater depth once the algorithms have been narrowed down.

References

[1] H. Markowitz, "Portfolio selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952. [Online]. Available: http://www.jstor.org/stable/2975974

[2] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters, "Noise dressing of financial correlation matrices," *Phys. Rev. Lett.*, vol. 83, pp. 1467–1470, Aug 1999. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.83.1467

[3] V. Tola, F. Lillo, M. Gallegati, and R. N. Mantegna, "Cluster analysis for portfolio optimization," *Journal of Economic Dynamics and Control*, vol. 32, no. 1, pp. 235–258, 2008, applications of statistical physics in economics and finance. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0165188907000462

[4] M. Huang, Q. Bao, Y. Zhang, and W. Feng, "A hybrid algorithm for forecasting financial time series data based on dbscan and svr," *Information*, vol. 10, no. 3, 2019. [Online]. Available: https://www.mdpi.com/2078-2489/10/3/103

[5] D. Wu, X. Wang, and S. Wu, "Construction of stock portfolios based on k-means clustering of continuous trend features," *Knowledge-Based Systems*, vol. 252, p. 109358, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950705122006815

[6] B. Laassem, A. Idarrou, L. Boujlaleb, and M. Iggane, "Label propagation algorithm for community detection based on coulomb's law," *Physica A: Statistical Mechanics and its Applications*, vol. 593, p. 126881, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0378437122000206

[7] S. Y. Yang, S. Y. K. Mo, and A. Liu, "Twitter financial community sentiment and its predictive relationship to stock market movement," *Quantitative Finance*, vol. 15, no. 10, pp. 1637–1656, 2015. [Online]. Available: https://doi.org/10.1080/14697688.2015.1071078

[8] A. Almog, F. Besamusca, M. MacMahon, and D. Garlaschelli, "Mesoscopic community structure of financial markets revealed by price and sign fluctuations," *PLOS ONE*, vol. 10, no. 7, pp. 1–16, 07 2015. [Online]. Available: https://doi.org/10.1371/journal.pone.0133679

[9] I. Anagnostou, T. Squartini, D. Kandhai, and D. Garlaschelli, "Uncovering the mesoscale structure of the credit default swap market to improve portfolio risk modelling," *Quantitative Finance*, vol. 21, no. 9, pp. 1501–1518, 2021. [Online]. Available: https://doi.org/10.1080/14697688.2021.1890807

[10] D. Kamuhanda, M. Cui, and C. J. Tessone, "Illegal community detection in bitcoin transaction networks," *Entropy*, vol. 25, no. 7, 2023. [Online]. Available: https://www.mdpi.com/1099-4300/25/7/1069

[11] S. M. Zema, G. Fagiolo, T. Squartini, and D. Garlaschelli, "Mesoscopic structure of the stock market and portfolio optimization," *Journal of Economic Interaction and Coordination*, vol. 20, no. 2, p. 307–333, Sep. 2024. [Online]. Available: http://dx.doi.org/10.1007/s11403-024-00426-y

[12] Wikipedia, "Hierarchical clustering — Wikipedia, the free encyclopedia," http://en.wikipedia.org/w/index.php?title=Hierarchical%20clustering&oldid=1309147899, 2025, [Online; accessed 07-September-2025].

[13] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[14] Wikipedia, "DBSCAN — Wikipedia, the free encyclopedia," http://en.wikipedia.org/w/index.php?title=DBSCAN&oldid=1296424268, 2025, [Online; accessed 07-September-2025].

[15] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2008, no. 10, p. P10008, oct 2008. [Online]. Available: https://dx.doi.org/10.1088/1742-5468/2008/10/P10008

[16] Wikipedia, "Louvain method — Wikipedia, the free encyclopedia," http://en.wikipedia.org/w/index.php?title=Louvain%20method&oldid=1298433126, 2025, [Online; accessed 07-September-2025].

[17] A. A. Hagberg, P. J. Swart, and D. S Chult, "Exploring network structure, dynamics, and function using networkx," *Proceedings of the 7th Python in Science Conference (SciPy2008)*, vol. 11, pp. 5–10, 2008.

[18] V. A. Traag and L. Šubelj, "Large network community detection by fast label propagation," *Scientific Reports*, vol. 13, no. 1, Feb 2023.

[19] Wikipedia, "Label propagation algorithm — Wikipedia, the free encyclopedia," http://en.wikipedia.org/w/index.php?title=Label%20propagation%20algorithm&oldid=1296650640, 2025, [Online; accessed 07-September-2025].

[20] R. Pardo, *Design, testing, and optimization of trading systems*, ser. Wiley Trader's Advantage S. Nashville, TN: John Wiley & Sons, Sep. 1992.

[21] A. Damani and N. Vaidya, "Is an equally weighted global investment portfolio the outperformer?" *Corporate Ownership and Control*, 2023. [Online]. Available: http://dx.doi.org/10.22495/cocv20i2art9

[22] A. Swade, S. Nolte, M. Shackleton, and H. Lohre, "Why do equally weighted portfolios beat value-weighted ones?" vol. 49, pp. 167 – 187, 2023. [Online]. Available: http://dx.doi.org/10.3905/jpm.2023.1.482

[23] M. Gelmini and P. Uberti, "The equally weighted portfolio still remains a challenging benchmark," *International Economics*, vol. 179, p. 100525, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2110701724000489